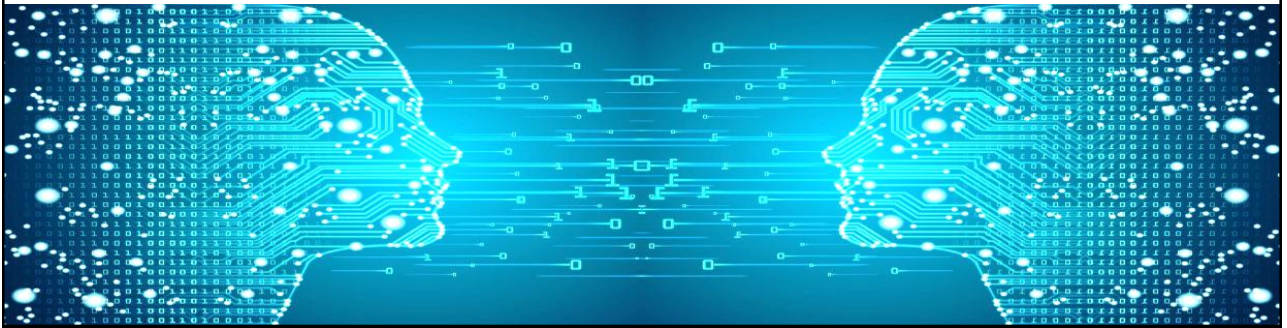
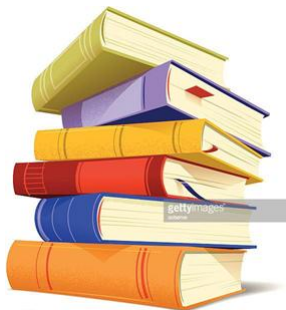


Machine Learning



An Introduction to Machine Learning



- Preliminaries
- Applications of Machine Learning algorithms
- The present and the future

Preliminaries



The Dream of Intelligent Machines

- Since their evolution, humans have utilized various tools to accomplish tasks, leading to the invention of machines that simplify life.
- However, intelligence remains the fundamental difference between humans and machines.
- Humans gather information using senses, analyze it, and make decisions. Machines, by nature, are not intelligent; they lack the ability to analyze data and make decisions.
- Example: A machine is not expected to understand "Harry Potter", jump over a hole, or form friendships.
- The era of intelligent machines began in the mid-twentieth century with Alan Turing's question: "Is it possible for machines to think?".
- This dream of creating machines with human-level intelligence has been expressed in many science fiction movies, such as Artificial Intelligence, The Matrix, and The Terminator.

AI Milestones (1/4)

- 1943: Warren McCulloch and Walter Pitts introduced the first neural network model.
- 1950: Alan Turing posed his famous question: "Can machines think?" and introduced the concept of a test of intelligence.
- 1955: Oliver Selfridge proposed using computers for pattern recognition.
- 1956: The term artificial intelligence was first used at the second Dartmouth conference, organized by John McCarthy, Marvin Minsky, Nathan Rochester of IBM, and Claude Shannon. The term cognitive science also originated in 1956.
- 1957: Frank Rosenblatt invented the first perceptron.

AI Milestones (2/4)

- 1959: John McCarthy invented the LISP programming language.
- 1962: David Hubel and Torsten Wiesel proposed neural networks for computer vision.
- 1964: Joseph Weizenbaum developed Eliza, the first expert system for disease diagnosis.
- 1969: Marvin Minsky and Seymour Papert's book Perceptrons highlighted limitations of neural networks, leading to a funding halt.
- 1979: Paul Werbos proposed the first efficient neural network model with backpropagation, rediscovered in 1986 by Rumelhart, Hinton, and Williams.

AI Milestones (3/4)

- 1987: Terrence Sejnowski and Charles Rosenberg developed NETTalk, an artificial neural network for speech recognition.
- 1987: John H. Holland and Arthur W. Burks invented an adapted computing system capable of learning.
- 1989: Dean Pomerleau proposed ALVINN (autonomous land vehicle in a neural network) for road following.
- 1997: IBM's Deep Blue chess machine defeated world chess champion Garry Kasparov.

AI Milestones (4/4)

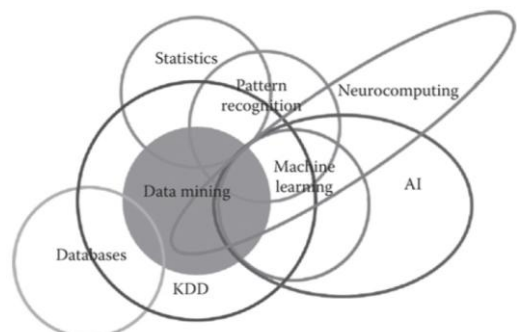
- 2011: IBM's Watson defeated Jeopardy! champions Brad Rutter and Ken Jennings.
- 1997-Present: Rapid developments in reinforcement learning, natural language processing, emotional understanding, computer vision, and computer hearing.
- Current Research Trends: Focus on computer vision, hearing, natural language processing, image processing and pattern recognition, cognitive computing, and knowledge representation.

Defining Machine Learning

- Learning is a highly personalized phenomenon for humans, involving curiosity, observation, and growth.
- However, learning is not exclusive to humans; even simple species like amoeba and paramecium, and even plants, exhibit intelligent behavior.
- The challenge is to introduce learning into human-made nonliving things, or machines.
- The term machine learning means to enable machines to learn without programming them explicitly.
- The main objectives of machine learning are to enable machines to make predictions, perform clustering, extract association rules, or make decisions from a given dataset.

Machine Learning: A Multidisciplinary Field (1/2)

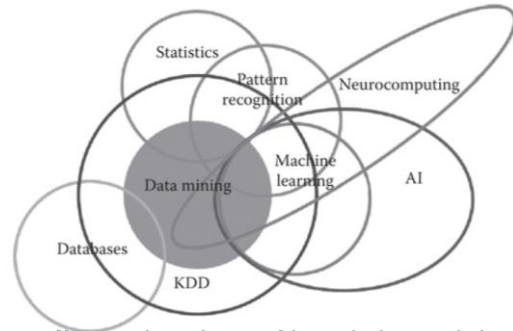
- Machine learning is a branch of artificial intelligence that aims to enable machines to perform tasks skillfully using intelligent software.
- Statistical learning methods form the backbone of this intelligent software.
- Machine learning algorithms require data to learn, thus connecting it with the discipline of database.
- It is closely related to terms like Knowledge Discovery from Data (KDD), data mining, and pattern recognition.
- Tom Mitchell's Definition (2006): Machine Learning is a natural outgrowth of the intersection of Computer Science and Statistics.



Different disciplines of knowledge and the discipline of machine learning

Machine Learning: A Multidisciplinary Field (1/2)

- Computer Science asks: "How can we build machines that solve problems?".
- Statistics asks: "What can be inferred from data plus a set of modeling assumptions, with what reliability?".
- Machine Learning builds on both, focusing on "how to get computers to program themselves (from experience plus some initial structure)" and "what computational architectures and algorithms can be used to most effectively capture, store, index, retrieve and merge these data".



Different disciplines of knowledge and the discipline of machine learning

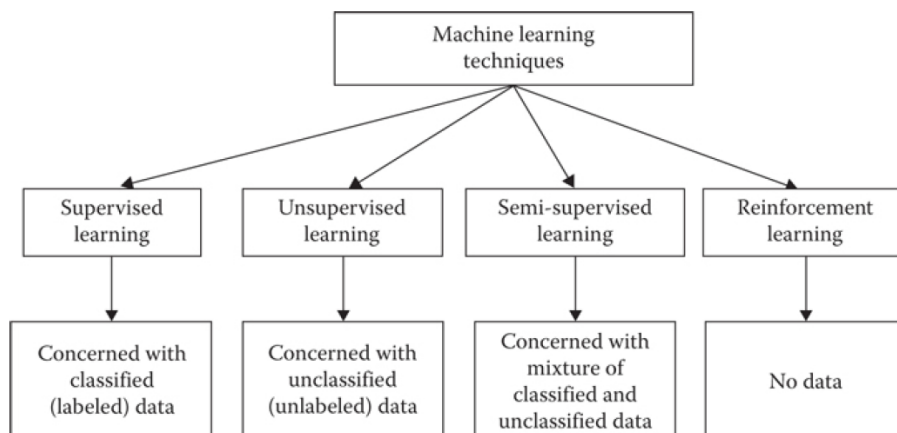
Why Machine Learning?

- Machine learning algorithms are helpful in bridging the gap of understanding how humans perform certain tasks effortlessly (e.g., speech recognition) but cannot explicitly explain.
- The idea is to write computer programs that make machines learn and perform tasks like prediction, without necessarily understanding the underlying human learning processes.
- The goal of learning is to construct a model that takes input and produces the desired result. This model can sometimes be understood, or it can be a "black box".
- Performance Measure: Besides speed and memory usage, the accuracy of results is a crucial measure for a machine learning algorithm.
- Tom Mitchell's Formal Definition of Learning (Carnegie Mellon University): "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."

Overview of Machine Learning Techniques (1/2)

- There are four general machine learning methods:
 - Supervised Learning
 - Unsupervised Learning
 - Semi-Supervised Learning
 - Reinforcement Learning
- This course focuses on supervised and unsupervised machine learning techniques.

Overview of Machine Learning Techniques (2/2)



Definition

- Definition: In supervised learning, the target is to infer a function or mapping from training data that is labeled.
- Training Data: Consists of an input vector X and an output vector Y of labels or tags.
- A label from vector Y explains its respective input example from input vector X .
- Together, X and Y form a training example.
- If labels do not exist for input vector X , then X is considered unlabeled data.
- Why "Supervised"? The output vector Y 's labels are provided by a supervisor, often humans, but sometimes machines.
- Human judgment is generally more expensive but yields higher accuracy compared to machine-labeled data. Manually labeled data is a precious and reliable resource for supervised learning.

Table 1.1 Unlabeled Data Examples along with Labeling Issues

<i>Unlabeled Data Example</i>	<i>Example Judgment for Labeling</i>	<i>Possible Labels</i>	<i>Possible Supervisor</i>
Tweet	Sentiment of the tweet	<i>Positive/negative</i>	Human/machine
Photo	Contains <i>house</i> and <i>car</i>	<i>Yes/No</i>	Human/machine
Audio recording	The word <i>football</i> is uttered	<i>Yes/No</i>	Human/machine
Video	Are weapons used in the video?	<i>Violent/nonviolent</i>	Human/machine
X-ray	Tumor presence in X-ray	<i>Present/absent</i>	Experts/machine

Example

- Scenario: Labeling unlabeled data based on different criteria.
- Table 1.1 (Unlabeled Data Examples along with Labeling Issues)
- Example Data: Email Text, Images, Speech, Customer Feedback, X-ray Image.
- Example Judgment for Labeling: Spam/Non-spam, Faces/Non-faces, Voice/Non-voice, Positive/Negative Sentiment, Tumor/Non-tumor.
- Possible Labels: Spam, Faces, Voice, Positive, Tumor.
- Supervisor Actor: Human or Machine (for first 4 cases), Expert Radiologist (for Tumor detection).
- Accuracy Concerns: While machines can be used for labeling in the first four cases, their low accuracy rates make human supervision preferable. Technologies like sentiment analysis, image recognition, and speech detection have progressed but still need improvement to match human performance.
- For critical tasks like tumor detection, normal humans cannot label X-ray data; expensive expert services are required.

Categories

- Two primary groups of algorithms fall under supervised learning:
 - Regression
 - Classification

Definition

- Definition: In unsupervised learning, supervisors or labeled training data are absent.
- Input: All we have is unlabeled data.
- Objective: The core idea is to find a hidden structure within this unlabeled data.
- Reasons for Unlabeled Data:
 - Unavailability of funds for manual labeling.
 - The inherent nature of the data itself.
 - Big Data Context: With numerous data collection devices, data is now collected at an unprecedented rate, characterized by variety, velocity, and volume.
- Extracting insights from this data without a supervisor is a significant challenge for machine learning practitioners today.

The Challenge

- The situation in unsupervised learning is likened to Alice's conversation with the Cheshire Cat in Alice's Adventures in Wonderland:
 - Alice: "Would you tell me, please, which way I ought to go from here?"
 - Cat: "That depends a good deal on where you want to get to."
 - Alice: "I don't much care where—"
 - Cat: "Then it doesn't matter which way you go,"
 - Alice: "—so long as I get somewhere,"
 - Cat: "Oh, you're sure to do that," said the Cat, "if you only walk long enough."
- In machine learning, clustering (an unsupervised learning algorithm) is analogous to the "walk long enough" instruction.
- Alice's "somewhere" is equivalent to finding regularities in the input data.

Definition

- **Definition:** This type of learning uses a mixture of classified (labeled) and unclassified (unlabeled) data to generate an appropriate model for data classification.
- **Typical Scenario:** Labeled data is scarce, while unlabeled data is abundant.
- **Objective:** To learn a model that can predict classes of future test data better than a model generated using labeled data alone.
- **Human Learning Analogy:** Our own learning process is similar to semi-supervised learning:
 - Children are supplied with unlabeled data from their environment.
 - They receive labeled data from a supervisor (e.g., a parent teaching names of objects).
- **Note:** This course will not discuss semi-supervised learning further.

Definition

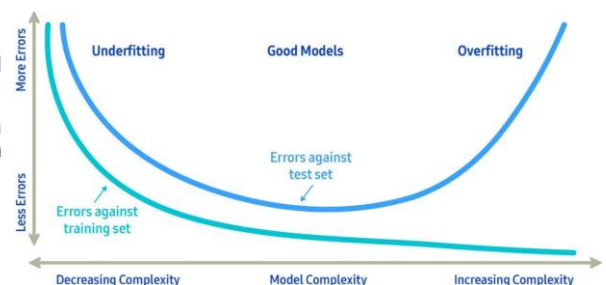
- **Definition:** The reinforcement learning method aims at using observations gathered from the interaction with the environment to take actions that would maximize the reward or minimize the risk.
- **Steps to Produce Intelligent Programs (Agents):**
 - Input state is observed by the agent.
 - A decision-making function is used to make the agent perform an action.
 - After the action, the agent receives reward or reinforcement from the environment.
 - State-action pair information about the reward is stored.
 - **Policy Fine-Tuning:** The stored information helps in fine-tuning the policy for a particular state in terms of action, leading to optimal decision making for the agent.
- **Note:** This course will not discuss reinforcement learning further.

Validation and Evaluation

- Purpose: Essential for assessing the quality of a model learned from a machine learning algorithm.
- Plato's Perspective on Stability of Belief: Plato discussed how true beliefs can "wander" if based on mistaken reasons, even if correct in current circumstances.
- Example: Identifying sheep based on features that don't distinguish them from goats might work in an environment without goats, but fails when goats are present, illustrating "counterfactual reliability".
- Machine Learning Analogy: If a function fits perfectly on training data, the machine learning community questions its performance on testing data.

Overfitting & Underfitting

- Overfitting
 - Overfitting occurs when the model generated tries to fit the training data too closely
 - The model is only good for the training data and will likely produce very poor results with any new test data
 - Basically useless
- Underfitting
 - Underfitting occurs when the model generated has too much deviation from a good predictive model
 - While this model may not be completely useless, the margin of error is too large
- The best machine learning models are simple models that fit the data well



Why Split Data?

- To evaluate model performance fairly
- To prevent overfitting
- To simulate unseen data scenarios
- To tune hyperparameters

How to split data

- The labeled data is splitted into three parts:
 - Training set: used to train/build the model
 - Validation set: used to tune hyperparameters
 - Test set: Used to validate the built model
- Typical Ratios (Train – Validation – Test):
 - 70% – 15% – 15% (Used when data set is medium/large)
 - 60% – 20% – 20% (Used when data set is small)
 - 80% – 20% (Train – Test, with cross-validation and small data set)

Cross-Validation

- Cross-validation is especially useful when the available training dataset is quite small and it is not possible to retain a portion of the data just for validation.
- k-Fold Cross-Validation:
 - Split data into k equal folds
 - Repeat k times → each fold used as validation once:
 - Train on **K-1 folds**
 - Validate on the **remaining fold**
 - Compute the **average performance** across all folds
- Ensures more reliable evaluation

Applications of Machine Learning algorithms



Applications of Machine Learning Algorithms

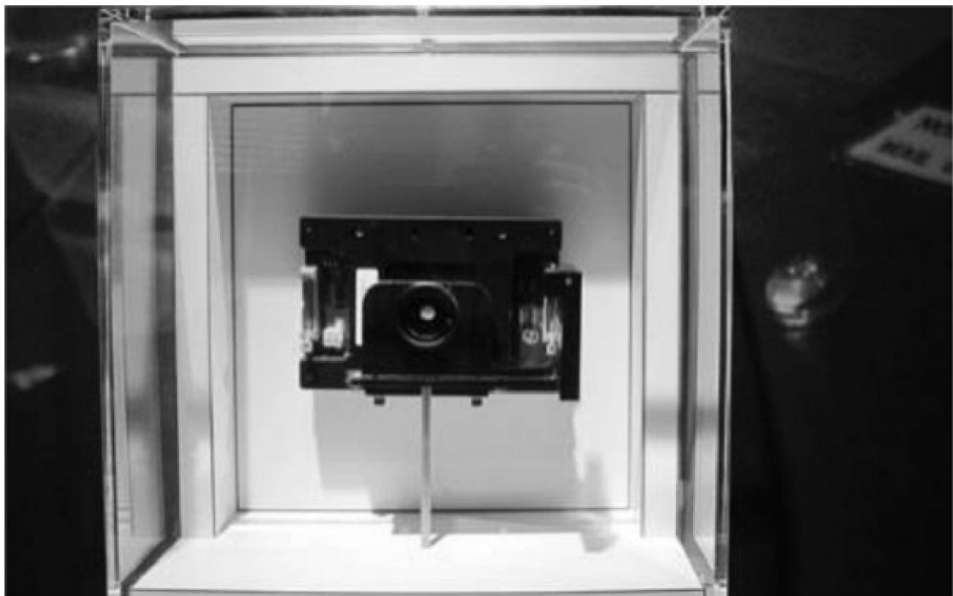
- Machine learning has proven to be the answer to many real-world challenges, though some problems still require breakthroughs.
- Bill Gates: "A breakthrough in machine learning would be worth ten Microsofts."
- In the following slides, we will discuss some key applications of machine learning.

Automatic Recognition of Handwritten Postal Codes

- Background: Postal services still exist despite digital communication. The US Postal Service successfully exploited machine learning potentials in the 1960s.
- They used machines to automatically read city/state/ZIP code lines of typed addresses to sort letters.
- Optical Character Recognition (OCR) Technology: Used machine learning algorithms to correctly interpret postal addresses.
- Definition: OCR technology makes image-based text content (typed, handwritten, printed) readable for machines.
- OCR software analyzes images to identify alphabetic letters and numeric digits, converting them into machine-encoded text.
- Machine-encoded text can be edited, searched, compressed, and used as input for applications like automatic translation, text-to-speech, and text mining.
- Accurate OCR makes data entry simpler, faster, and cheaper.

OCR - Impact and Evolution

- Scale of Postal Service: In 2006, the US Postal Service sorted and delivered over 213 billion pieces of mail, about 40% of the world's total. This "herculean service" is provided with the help of machines.
- OCR brought a revolution in postal system efficiency, connecting physical mail with information systems.
- Accuracy: According to the US Postal Service, machines read nearly 98% of hand-addressed letter mail and 99.5% of machine-printed mail.
- Google's OCR: Provides free service to convert text images to text documents for 200 languages in over 25 writing systems, using hidden Markov models and treating input as whole sequences.
- OCR is an application of pattern recognition, a branch of machine learning that focuses on recognizing patterns and regularities in data (text, speech, image).



An OCR camera used by the US Postal Service

Computer-Aided Diagnosis (CAD)

- Role: Pattern recognition algorithms in CAD assist doctors in interpreting medical images (X-rays, MRI, ultrasound) rapidly.
- Challenge: Radiologists must analyze and evaluate digital medical images under short time constraints, making machine assistance crucial.
- How it Works: CAD uses pattern recognition techniques from machine learning to identify suspicious structures in images.
- This is typically achieved through supervised learning, where thousands of labeled medical images are fed to machine learning algorithms (e.g., Bayesian classifier, artificial neural network, support vector machine).
- The resulting classifier is expected to correctly classify new medical images.

Computer-Aided Diagnosis – Risks (1/2)

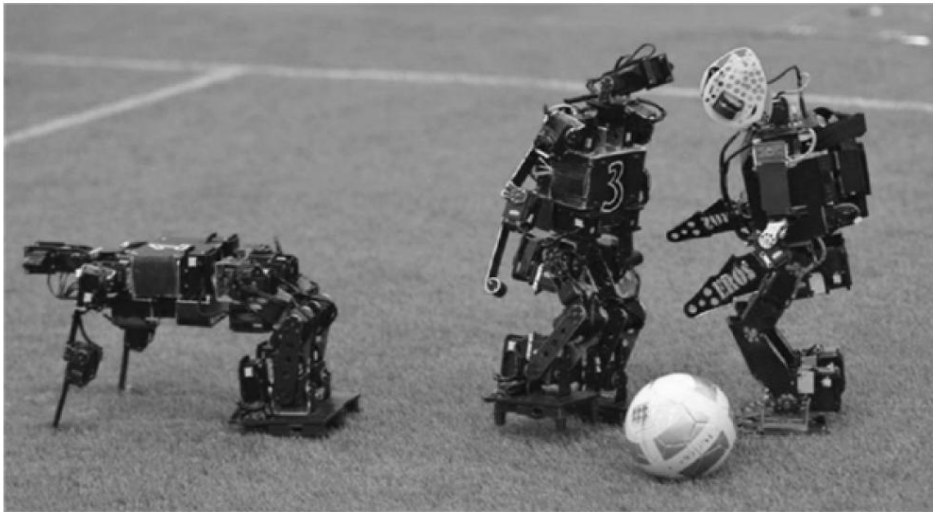
- Consequences of Mistakes: Mistakes in diagnosis by ML algorithms can have disastrous consequences, causing monetary damage and risking lives.
- False Positive: If a classifier wrongly detects breast cancer in a healthy patient:
 - Harmful psychological conditions for the patient.
 - Further confirmatory tests lead to monetary losses.

Computer-Aided Diagnosis – Risks (2/2)

- **False Negative:** If a classifier fails to detect breast cancer in a patient who actually has it:
 - Leads to wrong medical treatment.
 - Can threaten the patient's life in the near or far future.
- **Recommendation:** Complete substitution of doctors with technology is not recommended. The role of technology should be supportive, with the doctor (radiologist) taking responsibility for the final interpretation.
- **Examples of CAD Assistance:** Pathological brain detection, breast cancer, lung cancer, colon cancer, prostate cancer, bone metastases, coronary artery disease, congenital heart defect, Alzheimer's disease.

Computer Vision - RoboCup

- **Goal:** Enable robots to "see" and act appropriately by understanding their environment.
- Cameras provide images, but robots need pattern recognition to interpret them.
- RoboCup ("Robot Soccer World Cup"): An international robotic soccer tournament.
- **Official Goal:** "By the middle of the 21st century, a team of fully autonomous humanoid robot soccer players shall win a soccer game, complying with the official rules of FIFA, against the winner of the most recent World Cup."
- **Example:** RoboCup 2015 (China) attracted 175 intelligent robot teams from 47 countries.
- **ML in RoboCup:** Autonomous robots must cooperate, categorize objects, and recognize activities in adversarial dynamic environments. These tasks fall purely within the pattern recognition domain (a branch of machine learning), relying on camera input.



US and Iranian robot teams competing for RoboCup 2014 final

Computer Vision - Driverless Cars

- **Autonomous Cars:** A significant application where computer vision technology enables cars to operate without human drivers.
- **Industry Competition:** Major car companies (Toyota, General Motors, Nissan) and tech companies (Google, Tesla) are actively developing and testing driverless cars.
- **How they Work (Toyota's Example):**
 - Cars use multiple external sensors to recognize nearby vehicles and hazards.
 - They select appropriate routes and lanes based on the destination.
 - They "automatically operates the steering wheel, accelerator and brakes" to drive like a person.
- **Sensitivity:** Driverless car accidents can lead to tragedies. Similarly, computer vision in drones used in warfare can cause harm if algorithms misbehave.



Toyota tested its self-driving Highway Teammate car on a public road

Computer Vision - Face Recognition and Security

- Data Volume: Images from smartphones and CCTV cameras are generated at an unprecedented rate.
- Problem: Associating a face image with its respective identity.
- Challenge: Building a classifier for this task is non-trivial due to the vast number of classes and multiple image-related problems.
- Benefit for Security Agencies: Face recognition helps security agencies leverage large amounts of data from various sources to automatically perform tasks that are very difficult for humans to do manually.

Speech Recognition

- **Goal:** Develop methodologies and technologies to enable computers to recognize and translate spoken language into text.
- **Historical Impact:** Eliminates the need for stenography (writing in shorthand).
- **Applications:** Automatic transcription of speech into text is used in:
 - Video captioning
 - Court reporting
 - Assisting people with disabilities
- **Advancements:** The accuracy of speech recognition engines is continuously increasing.
- **Voice-Controlled Programs:** Examples include Apple's Siri, Google Now, Amazon's Alexa, and Microsoft's Cortana. While they don't always fully understand human speech, improvements are expected.

Text Mining

- **Data Source:** Text data is another crucial source of learning for machines.
- **Challenge:** How to use unstructured data or text, especially since most enterprise-related information is stored in text format.
- **Early Definition** (H.P. Luhn, 1958, IBM journal): Business intelligence systems "utilize data-processing machines for auto-abstracting and auto-encoding of documents and for creating interest profiles for each of the 'action points' in an organization."
- **Social Media:** Provides an abundance of unstructured text data, offering stakeholders (e.g., businesses) opportunities for analysis and beneficial use.

Text Mining – Applications

- Business intelligence
- National security
- Life sciences
- Sentiment classification
- Automated advertisement placement
- Automated classification of news articles
- Social media monitoring
- Spam filter

Text Mining – Combined Data Usage (1/2)

- It is possible to solve problems by combining both text and image data.
- Example 1: Author Identification for a Written Corpus:
 - Handwriting Detection: Using a known corpus of handwritten data to build a classifier that assigns a document to an author based on features.
 - Writing Style Detection (Text Mining Problem): Finding features related to a peculiar author from known documents to build a classifier that identifies document authorship.
 - Combined Approach: Joining the two classifiers can lead to improved performance for author identification.

Text Mining – Combined Data Usage (2/2)

- **Example 2: Identification of Unwanted Material in a Video:**
 - **Video Image Analysis:** Applying machine learning techniques on image data from video to build a model for identifying unwanted material.
 - **Social Media Comment Analysis:** Using comments related to the video from social media to understand video content and predict the presence or absence of unwanted material.
 - **Combined Approach:** The two classifiers can be combined to improve the overall system's performance.

The Present and the Future



Thinking Machines

- The question "whether a machine can think" was first asked by British mathematician Alan Turing in 1955, marking the start of artificial intelligence history.
- Turing also proposed a test to measure a machine's performance in terms of intelligence.

The Turing Test

- Basis: Alan Turing's 1950 question-and-answer game, "Can Machines Think?".
- Experiment: Investigates whether people can detect if they are talking to machines or humans.
- Passing Criteria: A computer passes the test if it is mistaken for a human more than 30% of the time during a series of five-minute keyboard conversations.
- Historic Achievement (2014): A computer program named Eugene Goostman passed the Turing test for the first time on June 7, 2014, at the Royal Society in London. Eugene convinced 33% of 30 human judges that it was human.

Chatbots - Capabilities and Limitations

- Definition: A chatbot is a computer program that simulates an intelligent conversation with one or more human users via audio or text.
- Loebner Prize: An annual competition where chatbots are judged for their human-likeness.
- 2015 Winner: Chatbot Rose: A judge noted that Rose (the chatbot) could handle simple questions ("where do you live," "what do you do") but "fell apart" when asked about complex or messy human topics (e.g., London house prices, dealing with slugs in the garden).
- Chatbot Strategy: They tried to divert the conversation or ignore complex questions.

Chatbots - Capabilities and Limitations

- Chatbot Strategy: They tried to divert the conversation or ignore complex questions.
- Conclusion by Judge: It will take time before a computer truly passes the Turing Test; humans are "much more interesting to talk to".
- Google's Advanced Chatbot: Programmed to learn from dialogue examples (IT helpdesk and movie transcripts) using a recurrent neural network.
- This means they offer more than "canned answers".
- Limitations acknowledged: still unable to have a realistic conversation, despite surprising answers to diverse questions without explicit rules.

Smart Machines - Definition

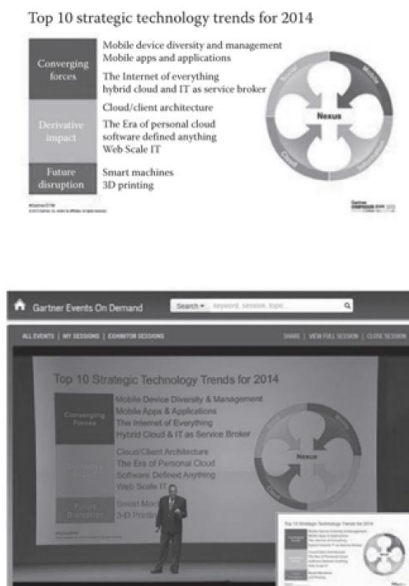
- The dream of machines appearing as smart as humans is still far from realized.
- General Definition: A smart machine is an intelligent system that uses equipment such as sensors, RFID, Wi-Fi, or cellular communications links to receive data and interpret it to make decisions.
- They utilize machine learning algorithms to accomplish tasks usually performed by humans to enhance efficiency and productivity.
- Gartner, Inc.'s Criteria for a True Smart Machine (Kenneth F. Brant):
- A smart machine does something that no machine was ever thought to be able to do.
- Example: A drone delivering a package.
- The machine is capable of learning.
- Note: The delivery drone, while smart, fails the "learning" test.

Smart Machines - Strategic Predictions

- Top 10 Strategic Technologies: Smart machines were predicted by Gartner, Inc. as strategic for most organizations in 2014 and 2015.
- 2014: Placed in the "future disruption" category alongside 3D printers.
- 2015: Included in the "intelligence everywhere" category.
- Predictions for Smart Machines:
- By 2015: More than 40 vendors with commercially available managed services leveraging smart machines.
- By 2018: Total cost of ownership for business operations reduced by 30% through smart machines.

Smart Machines - Strategic Predictions

- Through 2020: The smart machine era will "blossom" with intelligent personal assistants, smart advisors (like IBM Watson), advanced global industrial systems, and autonomous vehicles.
- The smart machine era will be the most disruptive in the history of IT.
- Well-known Examples: IBM's Watson, Google Now, and Apple's Siri.



The top 10 strategic technologies in years 2014 and 2015

IBM Deep Blue (1/2)

- **Historical Achievement:** In May 1997, Deep Blue became the first computer system to defeat a reigning chess world champion, Garry Kasparov, in a match.
- **Capabilities:** Utilized the "brute force of computing power" due to specialized hardware.
- **Capable of evaluating 200 million chess positions per second.**

IBM Deep Blue (2/2)

- **How it Evaluated:**
 - Deep Blue's evaluation function was initially written in a generalized form with many customizable parameters (e.g., importance of safe king position vs. space advantage).
 - Optimal values for these parameters were determined by the system itself through analyzing thousands of master games.
 - The evaluation function had 8,000 parts, many designed for special positions.
 - Its opening book contained over 4,000 positions and 700,000 grandmaster games.
- **Evolution:** In 1997, Deep Blue was a dedicated supercomputer. By 2006, chess programs like Deep Fritz (running on a personal computer) could evaluate 8 million positions per second, demonstrating improved software efficiency.

IBM Watson

- Name Origin: Named after IBM's first CEO, Thomas J. Watson.
- Capabilities: A "wonderful machine" capable of answering questions posed in natural language.
- Public Recognition: Gained worldwide fame by winning the first prize in the quiz show "Jeopardy!".
- Impact: With its supercomputing and AI power, Watson helps various industries by powering practical applications.
- Industries Benefiting: Healthcare, finance, legal, and retail sectors.
- Watson is considered "perhaps the most well-known example of artificial intelligence in use today".

Google Now

- Google's innovation, a "landmark for the machine learning world".
- A personal assistant with smartness and intelligence.
- Functions: Answering questions, making recommendations, and performing actions by assigning requests to web services.
- Users can use voice commands for reminders and trivia.
- Proactive Program: Observes user search habits to predict and deliver useful information proactively.

Apple Siri (Speech Interpretation and Recognition Interface)

- A widely used intelligent personal assistant by Apple Inc..
- Supports numerous languages (English, Spanish, French, German, Italian, Japanese, Korean, Mandarin, Russian, Turkish, Arabic).
- Continuously updated to improve responses.
- Context Understanding: Critical for Siri's effectiveness (e.g., reporting a threat instead of showing a map).

Microsoft Cortana

- Another intelligent personal assistant, competing with Google Now and Apple's Siri.
- Future integrations: Users will soon be able to use Skype to book trips, shop, and plan schedules by chatting with Cortana.

Conclusion

- Machine learning is a rapidly evolving field transforming various aspects of our lives.
- We've covered:
 - The core concept and history of machine learning.
 - Different types of learning: Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning.
 - Key applications from postal services to healthcare and autonomous vehicles.
 - The current state and future of AI, including famous intelligent systems.
 - The journey towards truly intelligent machines is ongoing, with significant breakthroughs and ethical considerations.

